

The Tao of Math

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The numbers you can count are not the real numbers

I like this Tao, because it weaves together a play on three different words

- Tao
- count
- real

Tao

Tao is an idea found in Chinese religions. From Wikipedia

Tao is considered to have ineffable qualities that prevent it from being defined or expressed in words.

and

The foundational text of Taoism, the Tao Te Ching, explains that Tao is not a 'name' for a 'thing' but the underlying natural order of the universe whose ultimate essence is difficult to circumscribe due to it being non conceptual yet evident in one's being of aliveness.

You get the idea. One of those opaque, mystical things that keeps the priests in business.

The Tao Te Ching itself begins

The Tao that can be spoken is not the eternal Tao.
The name that can be named is not the eternal name.

and this has led many commentators to coin similar statements of the form

The X you can Y is not the real X.

Some of these are meant seriously, as a guide to students of Tao, and some are meant humorously.

Real

Throughout most of history, numbers have been invented and used for practical purposes, by ordinary people rather than mathematicians. And those people generally only used the numbers that they needed.

The natural numbers

Probably the first numbers that anyone used were the natural numbers

1, 2, 3, 4, ...

These are also called the counting numbers, because of their obvious use in counting things, like sheep. Notably, shepherds didn't have zero, because if you don't got no sheep, you don't need a number to count the no sheep you

don't got.

Zero

Zero is an Arabic word that came to Europe when the Moors invaded Spain. (Many English words that start with z come from Arabic, like *zenith*). But the Arabs got zero (the idea; I don't know about the word) from the Hindus, and the Hindus needed zero because they wrote numbers with place-value notation, and when you write a number like

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then you do need a zero to show the no tens that you don't got, or else your tens get mixed up with your hundreds and then you don't know how many sheep you do got. But even shepherds who used place value notation didn't have negative numbers, because...well, because that just doesn't happen with sheep. But negative numbers did happen once the Italians got into banking during the Renaissance. In fact, the invention of negative numbers is sometimes credited to a particular Italian, who wrote something to the effect of

This problem cannot be solved unless it is conceded that the man began with a debt.

the point being that a debt is represented as a negative amount of money.

Imaginary numbers

Around the same time that Italian bankers were learning how to keep their books, Italian mathematicians were learning how to solve polynomial equations.

A linear equation has the form

$$ax + b = 0$$

and every linear equation has exactly one solution

$$x = -b / a$$

A quadratic equation has the form

$$ax^2 + bx + c = 0$$

and the solutions of a quadratic equation are given by the formulas

$$x = (-b + \sqrt{b^2 - 4ac}) / 2a$$

$$x = (-b - \sqrt{b^2 - 4ac}) / 2a$$

It looks like you get two solutions, but...there is a complication. That complicated bit in the middle

$$b^2 - 4ac$$

is called the *discriminant*. To get a solution, we have to take its square root

$$\sqrt{b^2 - 4ac}$$

and the problem is that you can't always take square roots. There are three cases

1. $b^2 > 4ac$

The discriminant is positive, so its square root is positive, and the two formulas give us two distinct solutions to the quadratic equation. Everybody is happy.

2. $b^2 = 4ac$

The discriminant is zero. The square root of zero is zero, so the two formulas both reduce to

$$x = -b / 2a$$

and we get a single solution to the quadratic equation. Some people say that we still get two solutions, but they are the same. Whatever.

3. $b^2 < 4ac$

The discriminant is negative. You can't take the square root of a negative number. Which is to say, negative numbers don't have square roots. The reason is that square numbers are never negative. If you square a positive number, you get a positive number

$$2 \cdot 2 = 4$$

If you square a negative number, you also get a positive number

$$-2 \cdot -2 = 4$$

So when the discriminant is negative, we *can't* take its square root, so we don't get any solutions to the quadratic equation. Bummer.

But, well, not every problem has a solution, and the Italians gamely moved on to cubic equations

$$ax^3 + bx^2 + cx + d = 0$$

Cubic equations are quite a bit more complicated than linear or quadratic equations. They also have fewer applications in everyday life, so they aren't commonly taught or studied by schoolchildren.

But they can be solved. Every cubic equation has at least one solution, and some have two, and some have three. And as the mathematicians were working their way through this, they discovered something very strange. Sometimes the formulas that you use to solve cubic equations call for you to take the square root of a negative number. Now, if you just stop right there because negative numbers don't have square roots, then you won't be able to evaluate the formulas, and you won't be able to solve the equation. But every cubic equation does have at least one solution, so what's the deal?

Eventually, someone decided to just, well...pretend. Pretend that you can take the square root of a negative number. You can't really, but we can write $\sqrt{-2}$, and pretend that it's a number, and treat it like a number.

And if you do this, and you keep doing what the formulas tell you to do, then eventually these pretend numbers go away, maybe because two of them get subtracted off

$$\sqrt{-2} - \sqrt{-2} = 0$$

so that there is nothing left, or maybe because two of them get multiplied together

$$\sqrt{-2} \cdot \sqrt{-2} = -2$$

so that you get a negative number back. And if you do this, then the formulas will give you the actual solutions to the original cubic equation.

It was obvious to the people who were using these pretend numbers that this whole thing was kind of a dodge, but it did let them solve their equations, so after a while they decided to just go with the flow and not worry about it too much.

Once these numbers were in use, they needed a name for them. They didn't call them *pretend* numbers. Maybe that was a little too undignified. Instead, they called them *imaginary* numbers. And then, naturally, they started calling those other numbers—numbers like 2 and 3 and 27—they started calling those numbers *real* numbers.

To the Renaissance mathematicians, the terms *real* and *imaginary* were not arbitrary names: they meant it. They regarded the real numbers as having some kind of actual existence that the imaginary numbers clearly lacked. I think there were two reasons for this.

The first is that Renaissance scholarship was still operating in a philosophical framework in which God created the universe, and things had essential natures, and Plato's ideal forms might really exist somewhere—maybe up there with God. Numbers like 1 and 2 and 3 were granted some kind of reality within this framework, while numbers like $\sqrt{-2}$ were regarded as merely the fevered imaginings of mathematicians.

The second is that they didn't know how to draw pictures of imaginary numbers. The real numbers are conventionally drawn on a number line, like

... -3 -2 -1 0 1 2 3 ...

There is obviously no place on this number line for something like $\sqrt{-2}$, so it can't really exist, right?

Eventually, people discovered that the imaginary numbers fit nicely on a perpendicular axis, like this

```

      .
      .
      .
    3i
    2i
     i
...-3 -2 -1 0 1 2 3 ...
    -i
   -2i
   -3i
      .
      .
      .

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Where i (short for imaginary) is the symbol for $\sqrt{-1}$, and then

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i =  $\sqrt{-1}$ 
2i =  $\sqrt{-4}$ 
3i =  $\sqrt{-9}$ 
etc...

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In this picture, the real numbers are on the horizontal axis, and the imaginary numbers are on the vertical axis. But it turns out that this isn't just an arbitrary convention. Once you've laid out the real and the imaginary numbers like this, you find that you can fill in the entire plane with numbers that are sums of real and imaginary numbers, sums like

$2 + 3i$

These are called complex numbers (uhhh...because they are more complex than either real or imaginary numbers alone) and the entire plane of these numbers is called the complex plane.

There are rules for doing arithmetic with complex numbers—adding, subtracting, multiplying, dividing—and the results that you get from that arithmetic correspond exactly to the positions of the complex numbers in the complex plane.

So in the end, the whole thing worked out beautifully, and today complex numbers are one of the foundational ideas in mathematics. But the Renaissance mathematicians didn't know any of this. They didn't even know how to draw the complex plane, so to them the imaginary numbers remained mysterious and opaque.

The modern view is that all numbers are real. Or maybe that all numbers are imaginary. Actually, the ontological status of numbers is still subject to some debate. Here is a Numberphile video that walks through a few current views on the question: [Do numbers EXIST?](#)

The real modern view is that we can have any numbers we like, and the criterion for liking numbers is that they should be interesting, which is to say, that they lead to interesting mathematics.

But despite the modern view (whatever that is) the terms *real* and *imaginary* persist to this day, and are universally understood to refer to those two kinds of numbers.

Count

Now we go back to the ancient Greeks and trace the development of numbers forward again, but different numbers this time.

Natural numbers

Again, the ancient Greeks had integers

1, 2, 3, ...

if only so that the shepherds could count their sheep.

Rational numbers

Perhaps surprisingly, the Greeks did not have fractions, like $1/2$ and $2/3$. But then, perhaps that's not so surprising. I mean, if you go out in the morning with 23 sheep, and when you bring them back in the afternoon you only count 22, so you go looking for your lost sheep, and what you finally find behind some rocks in the ravine is what the wolves left behind after they got it, then what you have is 22 sheep. Not 22 and $1/2$, not even 22 and $1/3$. Just...22. So, no fractions.

Actually, the Greeks did know how to work with fractions. They just didn't regard them as numbers. To the Greeks, $2/3$ wasn't a number, it was a ratio between two integers. They would make statements like

The lengths of lines A and line B are in the same ratio as the numbers 2 and 3.

The Greeks regarded the integers as having some kind of perfect or ideal existence, and they wanted to understand the entire world in terms of integers. Pythagoras (he of the famous theorem) was actually a cult leader of sorts. Their theology revolved around the integers; the whole thing had rather a numerological cast to it.

Irrational numbers

Of course, not everything in the world comes out in whole numbers, but the Pythagoreans assumed that quantities that were not integers themselves could still be understood as ratios of integers. Everything was going along nicely until someone proved that there was no ratio of integers corresponding to the ratio of the diagonal of a square to its side. In modern notation, we say that $\sqrt{2}$ is irrational (i.e. not a ratio of integers).

One way to think about the Pythagoreans is that they wanted the world to be simple. Neat. Clean. Integers are everything. And things that aren't integers are at least ratios of integers. And if there is some complicated, obscure thing that doesn't work out neatly in integers, well, hey, nothing's perfect, and maybe that will give us an interesting problem to work on.

But this wasn't something complicated and obscure. This was a square. One of the most basic geometric shapes. And finding out that a *square* could not be understood in terms of integers really upset the Pythagoreans. (Remember, they weren't just mathematicians: they were also a cult.) Legend has it that they took the guy who made this discovery out to sea and drowned him.

More prosaically, in *A Long Way From Euclid*, Constance Reid opines that the discovery that $\sqrt{2}$ is irrational was a turning point in the development of Greek mathematics. The Greeks didn't have a number for the length of the diagonal of a square, but they could still draw that diagonal, as a perfect, exact line segment, joining the opposite corners of a perfect, exact square. So they turned away from numbers and toward geometry.

Infinity

The integers go on for ever

1, 2, 3, 4, ...

and this leads to awkward questions about infinity. Like, what is infinity? How big is it? Is it a number? Does it exist? Can it exist? And so on. Like the questions about whether numbers are real, questions about infinity run through centuries of theology and philosophy. In *Wide Hats and Narrow Minds*, Stephen J. Gould writes

The standard example of ancient nonsense—the debate about angels on pinheads—makes sense once you realize that theologians were not discussing whether five or eighteen would fit, but whether a pin could house a finite or an infinite number.

Countable sets

There was very little advance on this debate until the 19th century, when a German mathematician named Georg Cantor found a way to make meaningful statements about the size of infinite sets. He started by formalizing what we do when we count things.

What we are really doing when we count things—he explained—is lining them up one-to-one with a list of numbers. If we have some things, like

* + # \$ %

and we want to know how many there are, we line them up with a list of the counting numbers, like this

*	+	#	\$	%
1	2	3	4	5

When everything is lined up nice and neat, we announce the top number in the list: there are *five* things in our set.

As long as you are dealing with finite sets, this seems obvious, or perhaps a distinction without a difference. But what Cantor realized is that you can do the same thing with infinite sets: try to line them up one-to-one. If you can, then we say that they have the same size. Here are some examples.

Suppose you're a shepherd, in Spain, and you have these numbers that you use to count your sheep

1, 2, 3, 4, ...

Sure, they're infinite, but, whatever.

Now the Moors appear, and they have all the same numbers you have, plus zero

0, 1, 2, 3, 4, ...

If we try to line up the two sets like this

1	2	3	4	...
1	2	3	4	... 0

well, that doesn't work at all. You can't tack an extra number onto the *end* of an infinite list: you'll never get there.

If we put it at the beginning

	1	2	3	4	...
0	1	2	3	4	...

then it looks like the Moors really do have one more number than we do: we've lined up all our counting numbers with all their counting numbers, and we still have the zero left over.

But we get to pick how we line up the numbers, and if we do it like this

1	2	3	4	5	...
0	1	2	3	4	...

then all of our numbers line up with all of their numbers nice and neat, and we see that the two sets of numbers have exactly the same size.

This is a little strange: it says that an infinite set can have the same size as part of itself. That doesn't happen with finite sets.

But it gets better. Now the Italians show up with their bankers and their credits and debits, and a whole extra infinity of negative numbers

..., -3, -2, -1, 0, 1, 2, 3, ...

Surely this is a bigger infinity, right? I mean, we can't push a whole infinity onto the front of our list: we'll never get to zero.

But, again, we don't have to line the numbers up in order: we just have to line them up somehow. And if we line them up like this

1	2	3	4	5	6	7	8	9	...
0	1	-1	2	-2	3	-3	4	-4	...

we see that the two sets do have the same size.

We can keep going. The ancient Greeks didn't admit fractions as numbers, but we do. Every fractional value on the number line: $1/2$ and $27/35$, and $298/3$ and...well...all of them. There's a lot. How are we going to line them up? The numerators go from 0 to infinity in both directions; the denominators go from 1 to infinity in both directions; we have an infinite number of infinite lists

⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮		
...	-5/5	-4/5	-3/5	-2/5	-1/5	0/5	1/5	2/5	3/5	4/5	5/4	...

...	-5/4	-4/4	-3/4	-2/4	-1/4	0/4	1/4	2/4	3/4	4/4	5/4	...
...	-5/3	-4/3	-3/3	-2/3	-1/3	0/3	1/3	2/3	3/3	4/3	4/4	...
...	-5/2	-4/2	-3/2	-2/2	-1/2	0/2	1/2	2/2	3/2	4/2	5/2	...
...	-5/1	-4/1	-3/1	-2/1	-1/1	0/1	1/1	2/1	3/1	4/1	5/1	...
...	-5/-1	-4/-1	-3/-1	-2/-1	-1/-1	0/-1	1/-1	2/-1	3/-1	4/-1	5/-1	...
...	-5/-2	-4/-2	-3/-2	-2/-2	-1/-2	0/-2	1/-2	2/-2	3/-2	4/-2	5/-2	...
...	-5/-3	-4/-3	-3/-3	-2/-3	-1/-3	0/-3	1/-3	2/-3	3/-3	4/-3	4/-4	...
...	-5/-4	-4/-4	-3/-4	-2/-4	-1/-4	0/-4	1/-4	2/-4	3/-4	4/-4	5/-4	...
...	-5/-5	-4/-5	-3/-5	-2/-5	-1/-5	0/-5	1/-5	2/-5	3/-5	4/-5	5/-5	...
.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.	.

But we can still do this. What we do is start in the middle, at 0/1, and work our way out in a spiral

—	.	.	.	.	.	.	.	.	.	.	.	.
...	-5/5	-4/5	-3/5	-2/5	-1/5	0/5	1/5	2/5	3/5	4/5	5/4	...
...	-5/4	-4/4	-3/4	-2/4	-1/4	0/4	1/4	2/4	3/4	4/4	5/4	...
...	-5/3	-4/3	-3/3	-2/3	-1/3	0/3	1/3	2/3	3/3	4/3	4/4	...
...	-5/2	-4/2	-3/2	-2/2	-1/2	0/2	1/2	2/2	3/2	4/2	5/2	...
...	-5/1	-4/1	-3/1	-2/1	-1/1	0/1	1/1	2/1	3/1	4/1	5/1	...
...	-5/-1	-4/-1	-3/-1	-2/-1	-1/-1	0/-1	1/-1	2/-1	3/-1	4/-1	5/-1	...
...	-5/-2	-4/-2	-3/-2	-2/-2	-1/-2	0/-2	1/-2	2/-2	3/-2	4/-2	5/-2	...
...	-5/-3	-4/-3	-3/-3	-2/-3	-1/-3	0/-3	1/-3	2/-3	3/-3	4/-3	4/-4	...
...	-5/-4	-4/-4	-3/-4	-2/-4	-1/-4	0/-4	1/-4	2/-4	3/-4	4/-4	5/-4	...
...	-5/-5	-4/-5	-3/-5	-2/-5	-1/-5	0/-5	1/-5	2/-5	3/-5	4/-5	5/-5	...
.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.	.

As we move along the spiral, we get a single list that has every fraction on it

0/1 -1/1 -1/-1 0/-1 1/-1 1/1 1/2 0/2 -1/2 -2/2 -2/1 -2/-1 -2/-2 ...

(It also has lots of duplicates, like 1/1, 2/2, 3/3, etc. You can strike these if you like.) That means that we can line the fractions up one-to-one with our counting numbers. And that means that the entire infinite-in-two-directions set of rational numbers is exactly the same size as our shepherd's original list of counting numbers.

One thing you'll notice is that when I made my list of fractions I didn't draw in the counting numbers, or the vertical lines that show the one-to-one correspondence. They aren't necessary. As long as you can make a list of the elements of a set, a list like

0/1	-1/1	-1/-1	0/-1	1/-1	1/1	1/2	0/2	-1/2	-2/2	-2/1	-2/-1	-2/-2	...
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then you can always line the counting numbers up with that list.

Sets that have this property—that you can make a list of them and line them up one-to-one with the counting numbers—are said to be *countable*, or sometimes *listable*. You can make a list of them. An infinite list, granted, but still a list, and every element of the set appears somewhere on the list. You can never count all of them—there's an infinite number—but if you are willing to count high enough, you can count to any given element of the set. That's what it means for a set to be countable.

The next set of numbers we want to look at is the algebraic numbers. Algebraic numbers are solutions of polynomial equations with integer coefficients. These were the kind of equations that the Italians were working on when they discovered imaginary numbers. They include all the integers, and all the rationals, plus a huge collection of irrational numbers. Here are some examples

equation	solution
$1x - 1 = 0$	1
$3x - 2 = 0$	$2/3$
$x^2 - 2 = 0$	$\sqrt{2}$
$x^3 - 47 = 0$	$\sqrt[3]{47}$

The exponents and the coefficients can go as high as we want: all the way to infinity. So, can we make a list of these numbers? Yes. Again, the trick is picking an order that gets every number on the list.

We start with the equations where no coefficient or exponent exceeds 1. There are only six of them

equation	solution
$-1x + -1 = 0$	-1
$-1x + 0 = 0$	0
$-1x + 1 = 0$	1
$1x + -1 = 0$	1
$1x + 0 = 0$	0
$1x + 1 = 0$	-1

Next we allow in the 2s

equation	solution
$1x + -2 = 0$	2
$1x + 2 = 0$	-2
$2x + -2 = 0$	1
$2x + -1 = 0$	$1/2$
$2x + 0 = 0$	0
$2x + 1 = 0$	$-1/2$
$2x + 2 = 0$	-1
$1x^2 + -2 = 0$	$\sqrt{2}, -\sqrt{2}$
$1x^2 + -1 = 0$	1, -1
$1x^2 + 0 = 0$	0
...	

yes, we have duplicates

and so on. I didn't finish the 2s. Even with just three numbers—0, 1, 2—there are a *lot* of these equations. But—and this is the crucial point—there are only a finite number equations that you can make if you only have 0, 1, and 2 to work with. So we can get all of them onto our list, and then move onto the equations that have 3s in them. And then the 4s, and then the 5s, and so on. So every algebraic equation appears *somewhere* on our list. And that means that every algebraic number also appears on our list: in the same place as the equation that it is the solution to.

So we can make a list of the algebraic numbers, and that means that the set of algebraic numbers—this enormous set, containing all the integers, and all the rationals, and a huge swath of irrational numbers besides—this set is exactly the same size as the shepherd's counting numbers. The algebraic numbers are countable.

When cantor worked all this out, he even made a name for the size of this set: he called it aleph-null. Aleph ($\aleph$) is the first letter of the Hebrew alphabet, and he added a zero to it, like a subscript, to indicate that this was the first (i.e. smallest) infinite number. People always pronounce it aleph-null, not aleph-zero, probably because Cantor was German, and the German word for zero is *null*.

Uncountable sets

OK, so we have some infinite sets, and we can show they are the same size, and we even made a name for that size. Is that it? Does this game of putting things in a list show that all infinite sets are the same size? Turns out not.

So let's go back to the ancient Greeks, who had integers, and knew how to work with ratios, even though they didn't consider fractions to be numbers. Then they discovered that $\sqrt{2}$ is not a ratio of integers, and had kind of a mathematical melt-down.

Some time in the 19th century, mathematicians finally figured out what the deal was with rational and irrational numbers. The rationals are dense on the number line. This means that between any two rational numbers, there is always another rational number. Which means, if you think about it, that between any two rational numbers there is an infinity of rational numbers. And if you keep thinking about it, your head starts to spin, but we don't need to go that far right now.

But for all that, there are, in an important sense, holes in the rational numbers. Holes like $\sqrt{2}$. Here's one way to think about it. $\sqrt{2}$ is not a rational number. That means that when you square a rational number, you never get 2. If you did, then the rational number that you squared would be $\sqrt{2}$, and that can't happen.

Therefore, when you square a rational number, the square is either less than 2 or greater than 2. Never 2 exactly. And that means that we can divide all the rationals into two groups: the ones that square to less than 2, and the ones that square to greater than 2.

What's more, these two sets are a bit strange. The lower set—the set that squares to less than 2—has no largest element. For any rational number that squares to less than 2, there is always another one, a little bit bigger, that also squares to less than 2. In the same way, the upper set has no smallest element. For any rational number that squares to more than 2, there is always another one, a little bit smaller, that also squares to more than 2. And right in-between the two sets, right where $\sqrt{2}$ would be, there is a hole where there is no rational number.

That hole is an irrational number. In fact, that hole is the *definition* of an irrational number. Any time you divide the rationals in this way—into a lower set and an upper set, where the lower set has no largest element and the upper set has no smallest element—you get a hole, which is an irrational number.

This construction of the irrational numbers was developed in the 19th century by the German mathematician Richard Dedekind, and a division of the rationals that produces an irrational number is called a *Dedekind cut*.

If we combine the rationals with the irrationals, then we get a new set that is the rationals with all those holes filled in. And then we can ask whether there are still holes in this new set of numbers. Are there things like $\sqrt{2}$ that aren't rational, and aren't irrational either, because all the rationals and irrationals are either less than or greater than, and never exactly equal to?

The answer is no. When you combine the rationals with the irrationals, the combined set has no holes in it. This was proved in the 19th century by the French mathematician Augustin-Louis Cauchy.

A set with this property—no holes—is called *complete*. In a sense, once you have a complete set, you have all the numbers. In particular, you have a number for every point on the number line.

The Renaissance Italians called the numbers on the number line *real* numbers, to distinguish them from imaginary numbers. Once the 19th century mathematicians proved that the combined set of rational and irrational numbers corresponds exactly to this real number line, they took to calling this combined set the set of real numbers.

So the word *real* now refers to two different distinctions

- real as opposed to imaginary
- real being the complete set of numbers, with no holes

The set of real numbers is sometimes called the continuum, the word *continuum* being another reference to the fact that the reals are complete, or continuous: they don't have any holes in them.

The proof that the reals are complete is not difficult. The hard part was understanding what irrational numbers are in the first place, and formulating a proper definition of them as holes in the rationals. That was the intellectual advance that took 2500 years from the time the Greeks realized that there were holes in the rationals to the time the Western Europeans figured out how to plug those holes with the irrationals.

There is an infinite number of irrational numbers. For example, the square root of any number that is not a perfect square is irrational

$\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, $\sqrt{6}$, ...

(But not $\sqrt{4}$. $\sqrt{4} = 2$.)

The reals include the rationals and the irrationals together. So there is an infinite number of reals. But is the set of reals any bigger than the set of counting numbers? Or can we find a clever way to make a list of the reals, thus showing that the size of the reals is aleph-null, the same size as every other infinite set we have looked at so far?

It turns out that we can not make a list of the reals. Cantor also proved this. Here's how the proof goes. The reals span the entire number line, from minus infinity to plus infinity. But let's set that aside and just consider the reals from 0 to 1. Suppose we had a list of these reals. What would it look like?

Every real number can be written as a decimal number with an infinite number of decimal places. The reals between 0 and 1 always begin 0.*something*. So the list looks like

0.0000000...
0.3333333...
0.5000000...
0.1124374...
0.4437584...
0.3495868...
0.5599433...
...

This is an infinite list of numbers, and each number extends off to the right with an infinite number of decimal places. I've put 0 first on the list (Why not? It has to be somewhere). And then 1/3, which is a repeating decimal, and then 1/2 which is called a terminating decimal, because all the decimal places after the 5 are 0, and then a bunch of numbers that I just made up, but they are still perfectly good real numbers, and they have to be somewhere on the list.

Now, if we look at that list, we can pull an interesting number out of it. Take the first decimal of the first number, the second decimal of the second number, the third decimal of the third number, and so on. We get

0.**0**000000...
0.3**3**33333...
...

```

0.5000000...
0.1124374...
0.4437584...
0.3495868...
0.5599433...
...

```

Or, putting all those digits on one line

```
0.0304563...
```

This is the number that you get by reading the digits down the diagonal in the list, and it is called a *diagonal* number. OK, maybe it's not that interesting. Although if our list really has all the reals between 0 and 1, then that diagonal number has to be on the list somewhere. It might be interesting to know where on the list it appears. I mean, if that sort of thing interests you. OK, it's not that interesting.

But now, like a magician, I'm going to pull a rabbit out of my hat. No, wait, that's a different trick. I'm going to pull a number out of my hat. A very interesting number. A very special number. A number that is **not** on the list. An anti-diagonal number. To pick an anti-diagonal number, we go down the diagonal, just like we did with the diagonal number, but instead of choosing each digit on the diagonal, we choose each digit to be **different** from the digit that is on the diagonal. Like

```

1
4
1
5
6
7
4
.
.
.

```

Or, on one line

```
0.1415674...
```

For simplicity, I just bumped each digit in the diagonal number up by one, but all that matters is that I pick a digit that is different from the one on the diagonal.

Now let's think about this number. It is a perfectly good real number, and it is between 0 and 1, so it has to be on our list somewhere. Is it the first number? No. It can't be the first number because the first number has 0 in the first decimal place, and the anti-diagonal number has 1 in the first decimal place. So it's not the first number. And it's not the second number, because the second number has 3 in the second decimal place, and the anti-diagonal number has 4 in the second decimal place. It can't be the third number, because of the 1 in the 3rd decimal place. It can't be the fourth number, because of the 5 in the fourth decimal place.

It can't be the Nth number, because it differs from the Nth number in the Nth decimal place. It differs from the Nth number in the Nth decimal place because that's how we constructed it. So it's not on our list.

When we try to line the reals up one-to-one with the counting numbers, we always have at least one real number left over. So we can't make a list of the reals. So the infinity of the reals is strictly greater than the infinity of the counting numbers. Because we can't make a list of the reals, we say that the reals are unlistable, and therefore uncountable.

When you first see this proof, it can be hard to appreciate how profound—how subversive—it is. All we have proven is that there is one real number that is not on our list. And we've been here before, right? The Moors

showed up with zero, and we just tacked it onto the front of our list. But we can't do that here. If we add our anti-diagonal number to the front of the list, then we get a new list, for which we can generate a new anti-diagonal number that is not on the new list.

So it's not just one number that's missing. It can't even be any finite number of reals that are missing—if it were, we would just tack them all onto the front of our list and be done.

So there is an infinity of reals missing from our list. And that infinity of missing reals isn't countable, either. If it were, then we would interleave the missing reals with the numbers on our list—the way we interleaved the negative integers with the positive integers—and be done.

So the reals are really, truly, uncountable. You can keep pulling countable infinities out of the reals and adding them to our list as long as you like, and what's left behind will always still be uncountably infinite.

The size of the reals is strictly greater than the size of the counting numbers. So now we have another infinite number; another aleph-something. Perhaps this is aleph-one, the next biggest infinite number. Or are there other infinite numbers, bigger than the counting numbers but smaller than the reals? The question of whether the reals are aleph-one is called the *continuum problem*, and amazingly, it remains an open question to this day.

There are definitely infinite sets that are bigger than the reals. In fact, there is an infinite number of infinite numbers, each bigger than the last. For example, there are an infinite number of curves in the plane, and that infinity is bigger than the infinity of the reals.

When Cantor first published this proof, it annoyed many people. He had proved that the infinity of the reals was greater—vastly, uncountably greater—than the infinities of the integers, the rationals, and the algebraics combined. Yet he had done this without producing a single one of these real numbers. It is a completely non-constructive proof.

Tao

Now that we know what the real numbers are, and what it means to count things, we can finally understand our Tao. The monks and the mystics are still meditating on their mountain tops, trying to scrue the inscrutable and eff the ineffable, but our Tao turns out to be simple and transparent, and absolutely true

The Tao of math: The numbers you can count are not the real numbers.

Notes

Whatever

Every quadratic equation

$$ax^2 + bx + c = 0$$

can be factored as

$$(x - r_1)(x - r_2) = 0$$

where r_1 and r_2 are the two roots. Even if r_1 and r_2 are the same, you still get two factors. So it's not whatever.

Algebraic numbers are solutions of polynomial equations with integer coefficients

Some people say *rational* coefficients. The two definitions are obviously equivalent: integers are rationals, and an equation with rational coefficients can be converted to one with integer coefficients (and the same roots) simply by multiplying through by all the denominators.

I suppose you say *integer* to emphasize the economy with which the algebraics can be generated, and *rational* to emphasize the size of a set which nonetheless turns out to be no bigger than the natural numbers.

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